GLS UNIVERSITY

Faculty of Computer Applications & Information Technology

Integrated Master of Computer Applications [iMSc.IT] Programme

Semester VIII Project Report for

221601801 FULL STACK DEVELOPMENT

"Bus Pass Management System"

Submitted by:

Group No.:

Hardik Sanjaybhai Parmar - 202201619010054

Dhairy Pradipbhai Makadia - 202201619010152

GLS UNIVERSITY

Faculty of Computer Applications & IT iMSc.IT Programme

Ahmedabad

CERTIFICATE

This is to certify that

1. Hardik Sanjaybhai Parmar

2. Dhairy Pradipbhai Makadia

Students of Semester-VIII iMSc.IT, FCAIT, GLS University have successfully completed the Project of 221601801 FULL STACK DEVELOPMENT as a part of their curriculum for the academic year 2025-2026.

Date of Submission: Project Guide - Name & Sign

TABLE OF CONTENTS

1. Introduction

I. Project Objective

II. Purpose

III. Modules

2. System Architecture

I. Technology Stack

II. Architecture Diagram

III. Project Structure

3. Features & Functionality

I. User Features

II. Admin Features

III. Authentication & Security

4. API Endpoints

5. Database Schema Design

6. Business Logic

I. Fare Calculation

II. Bus Stops

III. Pass Lifecycle

IV. QR Code Verification

7. Screenshots - User Side

8. Screenshots - Admin Side

9. Conclusion

1. INTRODUCTION

I. Project Objective

The Bus Pass Management System is a comprehensive full-stack web application designed to digitize and streamline the process of issuing, managing, and verifying bus passes. The system replaces traditional paper-based bus pass systems with a modern, efficient, and secure digital solution that benefits both commuters and transport authorities.

The primary objective is to create a seamless experience for users to purchase bus passes online, enable administrators to manage pass approvals efficiently, and provide conductors with real-time QR-based verification capabilities. The system supports individual and family pass purchases with dynamic fare calculation based on distance between bus stops.

II. Purpose

The purpose of this system is to modernize public transportation services by providing:

- Digital bus passes with QR code verification
- Online application and approval system
- Family pass management with auto-account creation
- Real-time pass tracking and status updates
- Role-based access for Admin, Superadmin, and Conductor

This system reduces waiting time at bus pass counters and ensures a smooth and secure experience for users and administrators.

III. Modules

1. User Authentication Module

- User registration and login with form validation
- Secure authentication using JWT (JSON Web Tokens)
- Password encryption using Bcrypt
- Forgot and reset password functionality via email

2. Individual Pass Module

- Select source and destination from 20 bus stops
- Dynamic fare calculation using Haversine formula

- Choose pass type (Monthly, Quarterly, Half-Yearly, Yearly)
- Submit pass request for admin approval

3. Family Pass Module

- Add multiple family members (up to 10)
- Auto-create user accounts for family members
- Purchase passes for entire family in one transaction
- 3-step wizard: Add Members, Route & Pass, Review

4. Dashboard Module

- View all passes with status (Active, Pending, Expired, Rejected)
- Display QR codes for active passes
- Pass details with verification code
- Responsive design for mobile and desktop

5. Admin Module

- Approve or reject pending pass applications
- View passes filtered by status
- Statistics dashboard (Total, Active, Pending counts)
- Superadmin can manage other admin accounts

6. Verification Module

- QR code scanning using device camera
- Manual verification using 16-character hex code
- Real-time validation with pass details display

2. SYSTEM ARCHITECTURE

I. Technology Stack

Component	Technology
Frontend (User)	React.js 19.x + Tailwind CSS + Vite
Frontend (Admin)	React.js 19.x + Tailwind CSS + Vite
Backend	Node.js + Express.js 5.x
Database	MongoDB Atlas (Mongoose 9.x ODM)
Authentication	JSON Web Token (JWT)
Password Security	Bcrypt.js
QR Code	qrcode library + html5-qrcode scanner
Email Service	Nodemailer (Gmail SMTP)
Routing	React Router 7.x
Deployment	Render.com (Cloud Platform)

II. Architecture Diagram

```
+---------------------+     +---------------------+
|   User Frontend     |     |   Admin Frontend    |
|   (React + Vite)    |     |   (React + Vite)    |
|   Port: 5173        |     |   Port: 5174        |
+----------+----------+     +----------+----------+
           |                           |
           +----------- + -------------+
                        |
              +---------v---------+
              |  Express.js API   |
              |  (Node.js)        |
              |  Port: 8000       |
              |  JWT Auth + CORS  |
              +---------+---------+
                        |
              +---------v---------+
              |   MongoDB Atlas   |
              |   Collections:    |
              |   users, admins,  |
              |   buspasses       |
              +-------------------+
```

III. Project Structure

```
bus-pass-management-system/
|-- backend/
|   |-- controllers/     # Business logic handlers
|   |-- models/          # Mongoose schemas
|   |-- routes/          # API route definitions
|   |-- middleware/      # Auth guards, HTTP logging
|   |-- utils/           # Token, QR, email, logger
|   |-- index.js         # Express server entry point
|
|-- frontend/            # Admin Portal (React + Vite)
|   |-- src/auth/        # Login, Register, auth service
|   |-- src/pages/       # Admin Dashboard
|
|-- user-frontend/       # User Portal (React + Vite)
|   |-- src/auth/        # Login, Register, ForgotPassword
|   |-- src/pages/       # Dashboard, FamilyPass, Profile
|   |-- src/components/  # Navbar, Footer
|   |-- src/layout/      # RootLayout wrapper
```

3. FEATURES & FUNCTIONALITY

I. User Features

- User Registration with name, email, phone (10-digit), DOB, and strong password validation
- JWT-based login with 1-day token expiry
- Email-based password reset with 6-character code (15-min expiry)
- Individual pass purchase with real-time fare display for 4 pass types
- Family pass purchase for up to 10 members with auto-account creation
- View pass history with QR codes, verification codes, and validity dates
- Responsive design for desktop and mobile devices

II. Admin Features

- Pass approval/rejection workflow with QR code generation on approval
- Camera-based QR scanner for real-time pass verification
- Manual verification using 16-character hex code
- Pass filtering by status (Pending, Active, Rejected, All)
- Role-based access: Admin (manage passes), Conductor (verify only), Superadmin (full control)
- Dashboard statistics: Total Passes, Active, Pending counts

III. Authentication & Security

- JWT token-based stateless authentication with 1-day expiration
- Bcrypt password hashing with salt rounds
- Role-based middleware: protectUser, protectAdmin, adminOnly, superAdminOnly, anyAdminRole
- CORS whitelisting for development and production origins
- Server-side input validation for all endpoints

4. API ENDPOINTS

User Authentication API

Method	Endpoint	Auth	Description
POST	/api/user/register	Public	Register new user
POST	/api/user/login	Public	User login, returns JWT
POST	/api/user/forgot-password	Public	Send reset code to email
POST	/api/user/reset-password	Public	Reset password with code
GET	/api/user/profile	User	Get user profile
PUT	/api/user/change-password	User	Change password

Bus Pass API (User)

Method	Endpoint	Auth	Description
GET	/api/user/bus-pass/stops	Public	Get 20 bus stops list
GET	/api/user/bus-pass/fare	Public	Calculate fare for route
POST	/api/user/bus-pass/buy	User	Purchase individual pass
POST	/api/user/bus-pass/buy-family	User	Purchase family passes
GET	/api/user/bus-pass/my-passes	User	Get user's passes
POST	/api/user/bus-pass/verify	Public	Verify pass (QR/code)

Admin API

Method	Endpoint	Auth	Description
POST	/api/admin/login	Public	Admin login
POST	/api/admin/register	Public	Register admin
GET	/api/admin/profile	Admin	Get admin profile
GET	/api/admin/bus-passes	Admin	Get passes by status
PATCH	/api/admin/bus-passes/:id/approve	Admin	Approve pass
PATCH	/api/admin/bus-passes/:id/reject	Admin	Reject pass
POST	/api/admin/bus-passes/verify	Any	Verify pass

5. DATABASE SCHEMA DESIGN

User Collection

Field	Type	Constraints	Description
_id	ObjectId	Primary Key	Auto-generated unique ID
name	String	Required	User's full name
email	String	Required, Unique	Email address
phone	String	Required	10-digit phone number
dateOfBirth	Date	Required	Date of birth
password	String	Required	Bcrypt hashed password
isActive	Boolean	Default: true	Account status

Admin Collection

Field	Type	Constraints	Description
_id	ObjectId	Primary Key	Auto-generated unique ID
name	String	Required	Admin's full name
email	String	Required, Unique	Email address
password	String	Required	Bcrypt hashed password
role	String	Enum	admin / superadmin / conductor
isActive	Boolean	Default: true	Account status

BusPass Collection

Field	Type	Constraints	Description
_id	ObjectId	Primary Key	Auto-generated unique ID
user	ObjectId	Ref: User	Pass owner reference
passType	String	Enum	monthly/quarterly/half-yearly/yearly
price	Number	Required	Calculated fare amount
source	String	Required	Source bus stop
destination	String	Required	Destination bus stop
validFrom	Date	On approval	Pass start date

validTill	Date	Calculated	Pass expiry date
qrToken	String	Generated	JWT QR token
qrImage	String	Generated	QR code data URL
code16	String	Unique	16-char hex verification code
memberName	String	Family	Family member name
memberEmail	String	Family	Family member email
memberRelation	String	Enum	Relation to user
status	String	Enum	pending/active/rejected/expired

6. BUSINESS LOGIC

I. Fare Calculation

The system uses the Haversine formula to calculate geographic distance between two bus stops using latitude and longitude coordinates. A road factor of 1.3x is applied for actual road distance.

- Base Rate: Rs. 3 per kilometer
- Minimum Fare: Rs. 200
- Monthly Fare = max(distance x 3, 200)
- Quarterly = Monthly x 2.4 (90 days)
- Half-Yearly = Monthly x 4.0 (180 days)
- Yearly = Monthly x 7.0 (365 days)

II. Bus Stops

The system supports 20 bus stops across Gujarat:

#	Stop Name	#	Stop Name
1	Ahmedabad (Paldi)	11	Navsari
2	Ahmedabad (Maninagar)	12	Valsad
3	Ahmedabad (Naroda)	13	Vapi
4	Gandhinagar	14	Rajkot
5	Nadiad	15	Jamnagar
6	Anand	16	Junagadh
7	Vadodara	17	Bhavnagar
8	Bharuch	18	Morbi
9	Surat	19	Mehsana
10	Palanpur	20	Himmatnagar

III. Pass Status Lifecycle

```
User Buys Pass         Admin Reviews
+----------+    +----------+----------+
|          |    |          |          |
| PENDING  +--->| APPROVE  |  REJECT  |
|          |    | (QR Gen) |          |
+----------+    +-----+----+----+-----+
```

```
      |         |
      v         v
  ACTIVE    REJECTED
    |
    v
 EXPIRED (past validTill)
```

IV. QR Code Verification

When a pass is approved, a JWT token is generated containing pass ID, user ID, and expiry date. This token is encoded into a QR code image. A unique 16-character hexadecimal code is also generated for manual verification. Conductors can verify using either the QR scanner or the code.

7. SCREENSHOTS - USER SIDE

User Portal URL: http://localhost:5173

Production URL: https://bus-pass-management-system-user-frontend.onrender.com

Screenshot: User Login Page

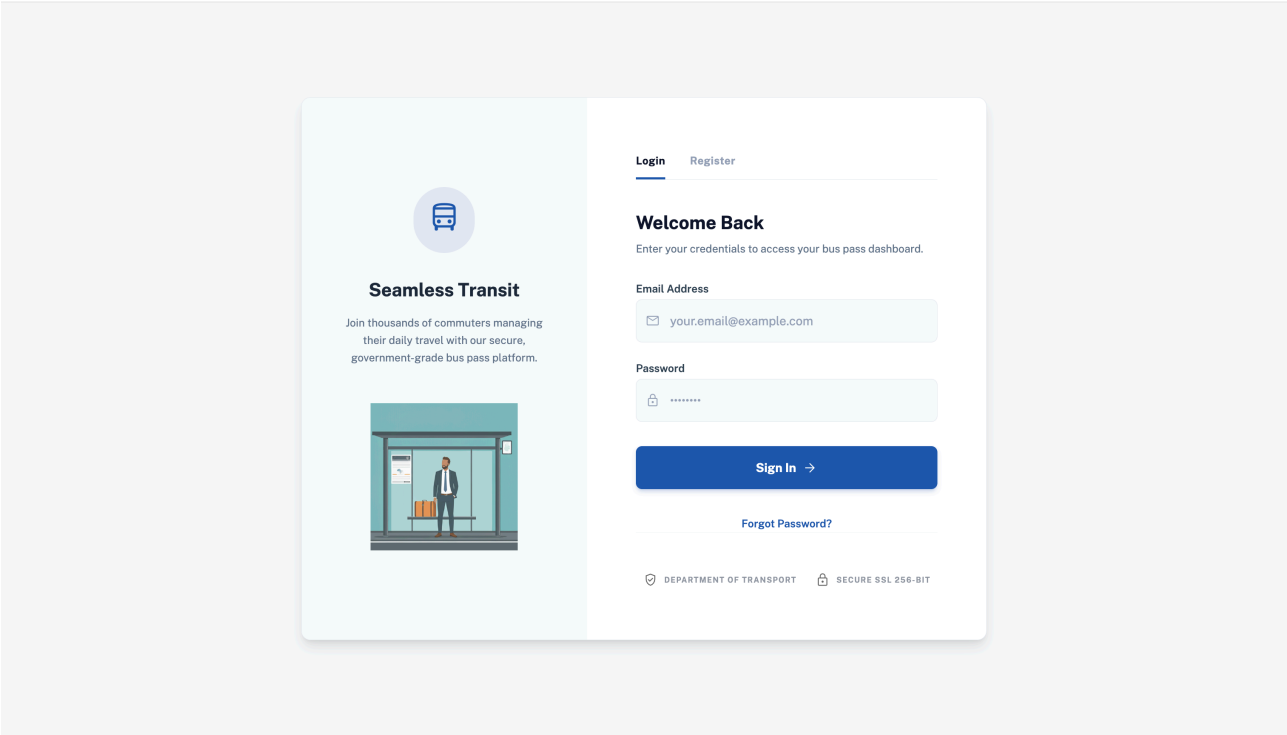

Figure: User Login Page - The login page allows registered users to sign in using their email and password. Features include form validation, a 'Forgot Password' link, and a link to the registration page. URL: http://localhost:5173/login

Screenshot: User Registration Page

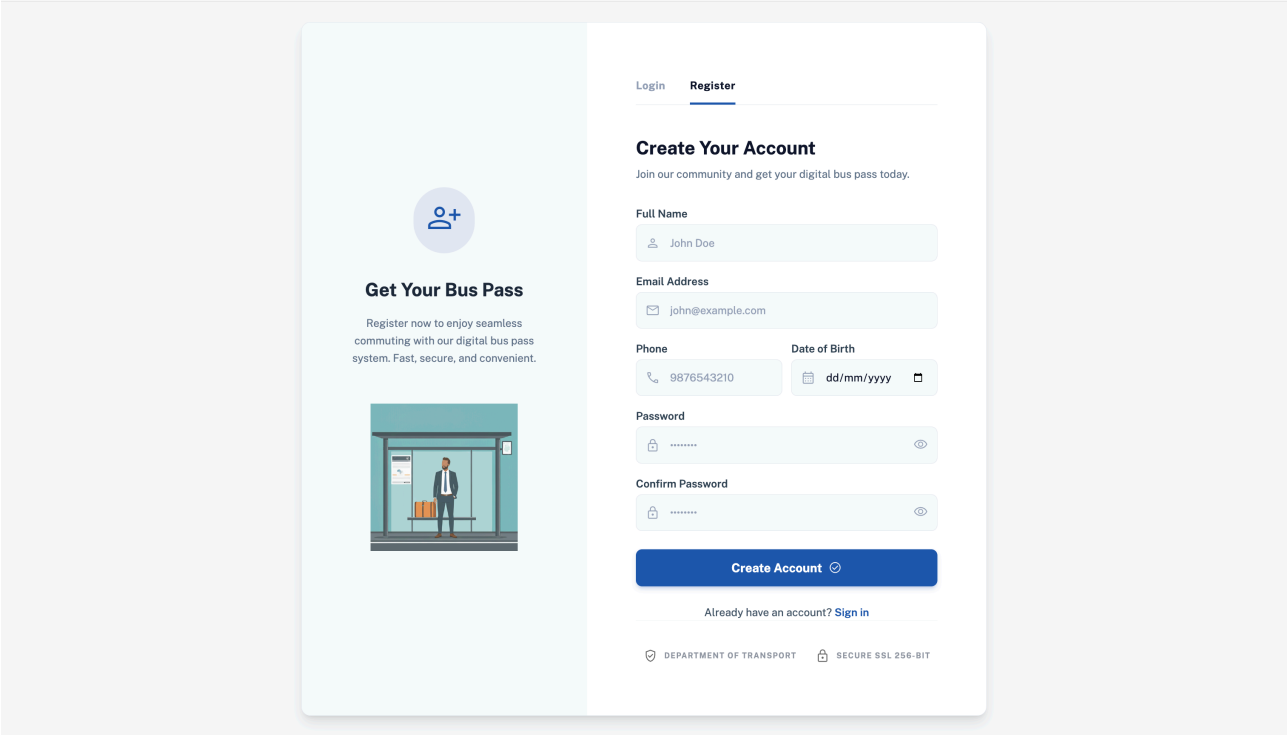

Figure: User Registration Page - New users can create an account by providing their full name, email address, phone number, date of birth, and a strong password with confirmation. URL: http://localhost:5173/register

Screenshot: Forgot Password Page

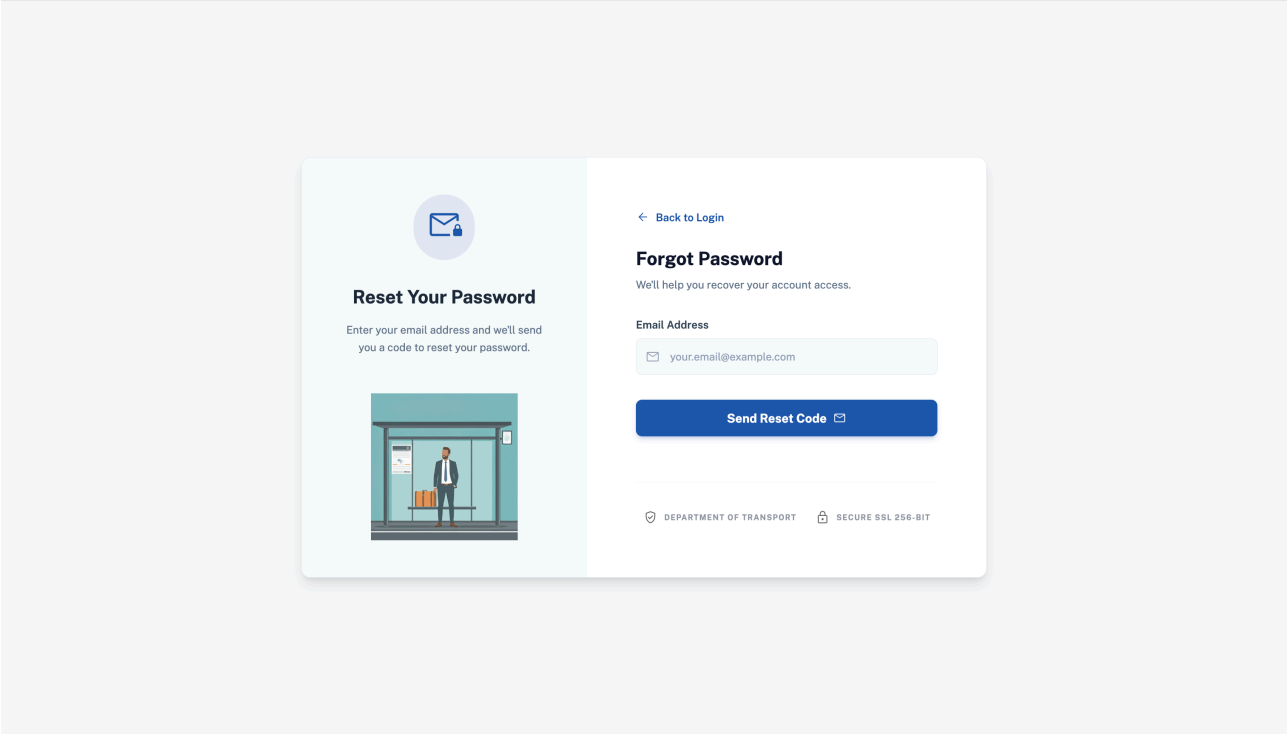

Figure: Forgot Password Page - Users can request a password reset code sent to their registered email. The page accepts an email address and sends a 6-character alphanumeric code valid for 15 minutes. URL: http://localhost:5173/forgot-password

Screenshot: User Dashboard - Buy a Bus Pass

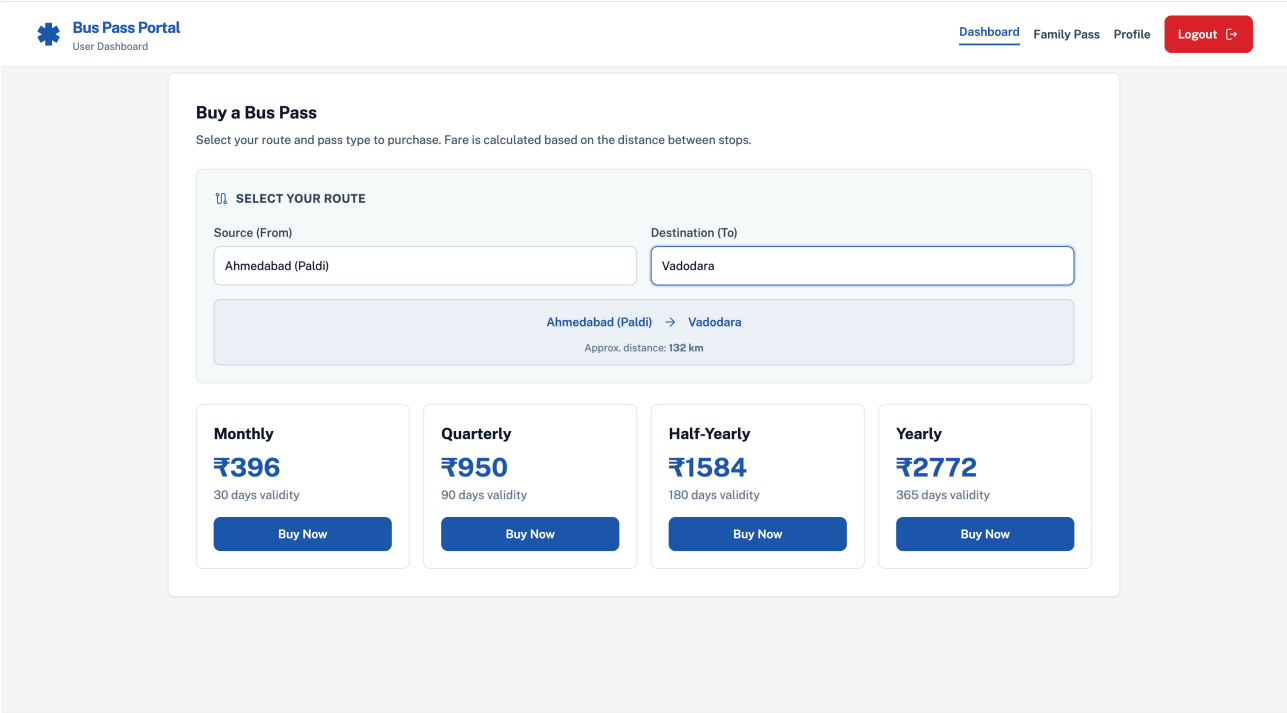

Figure: User Dashboard - Buy a Bus Pass - The main dashboard displays source and destination dropdowns with 20 bus stops, real-time fare calculation for all pass types (Monthly, Quarterly, Half-Yearly, Yearly), and Buy Now buttons. URL: http://localhost:5173/dashboard

Screenshot: User Dashboard - Active Pass with QR Code

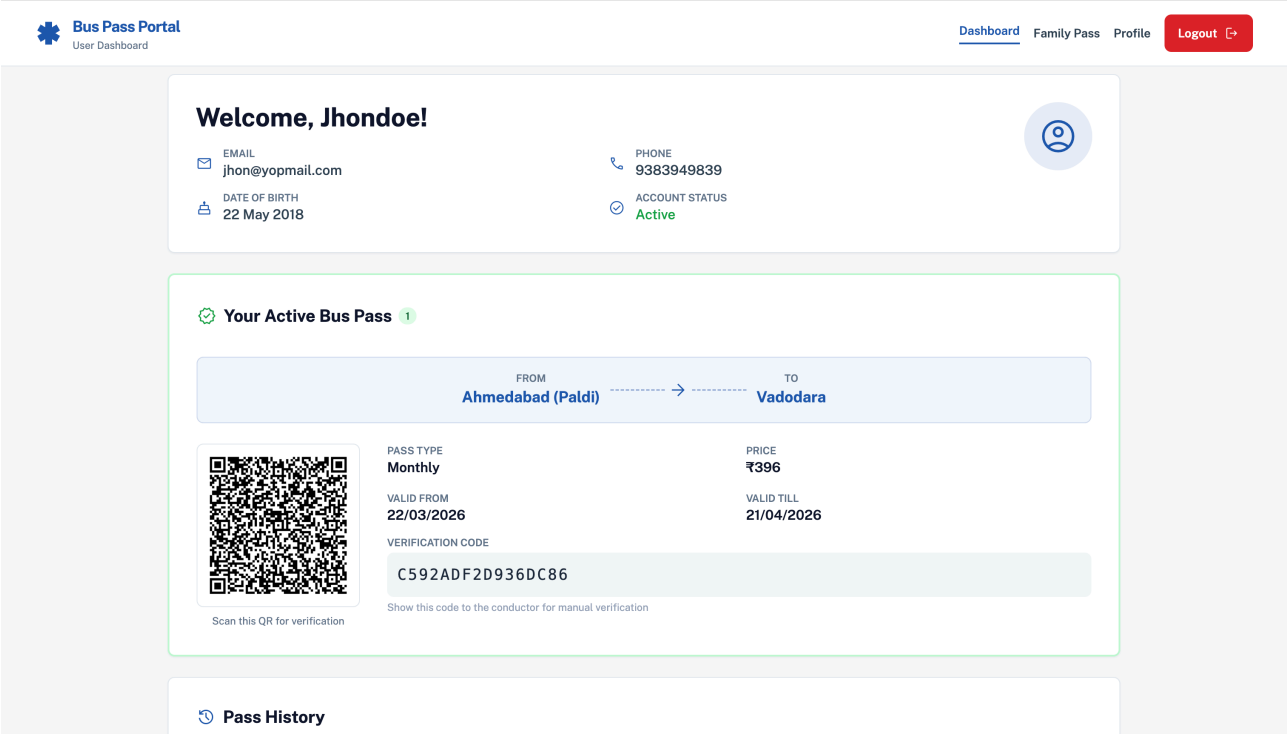

Figure: User Dashboard - Active Pass with QR Code - After admin approval, the dashboard shows the active bus pass with QR code for scanning, verification code for manual verification, validity dates, route details, and pass type

information. URL: http://localhost:5173/dashboard

Screenshot: Family Pass - Add Members (Step 1)

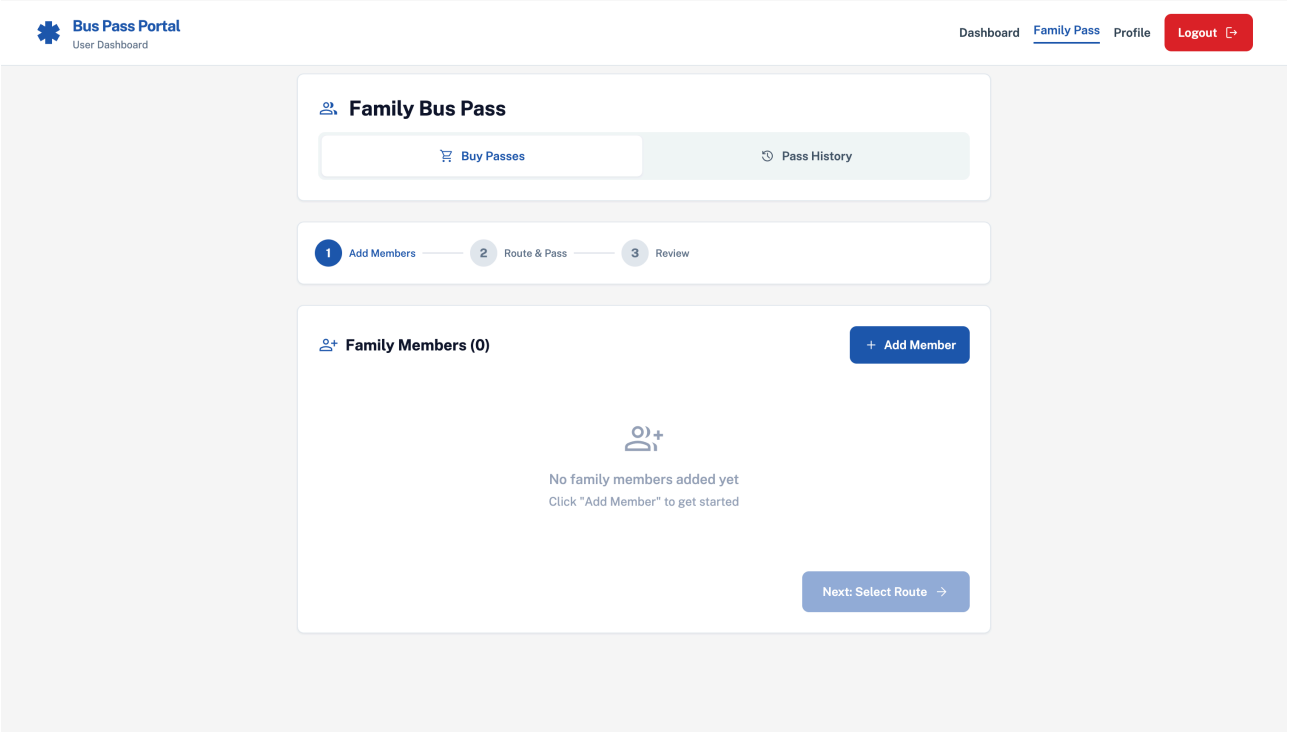

Figure: Family Pass - Add Members (Step 1) - Step 1 of the family pass purchase flow. Users can add multiple family members with the '+ Add Member' button. Shows a stepper navigation with 3 steps: Add Members, Route & Pass, Review. URL: http://localhost:5173/family-pass

Screenshot: Family Pass - Add Family Member Dialog

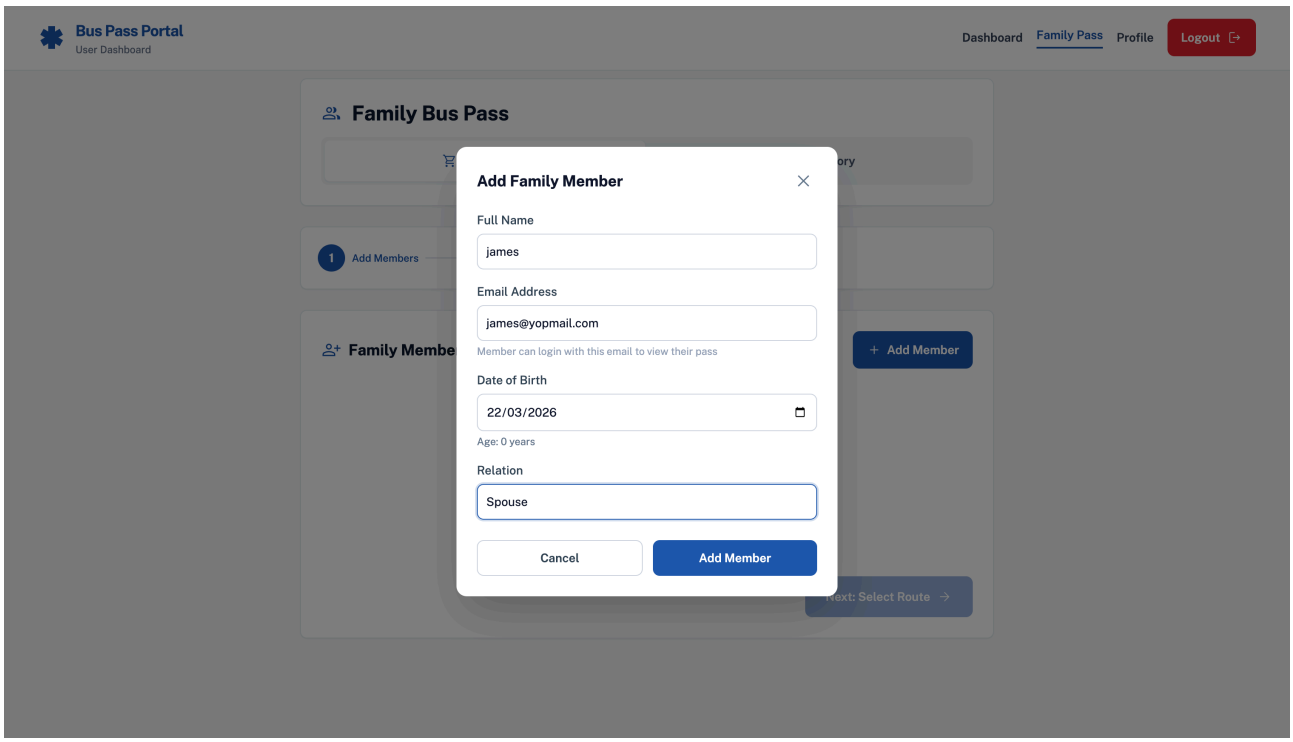

Figure: Family Pass - Add Family Member Dialog - A modal dialog for adding a family member with fields for Full Name, Email Address, Date of Birth, and Relation (Spouse, Father, Mother, Son, Daughter, etc.). URL: http://localhost:5173/family-pass

Screenshot: Family Pass - Member Added (Step 1)

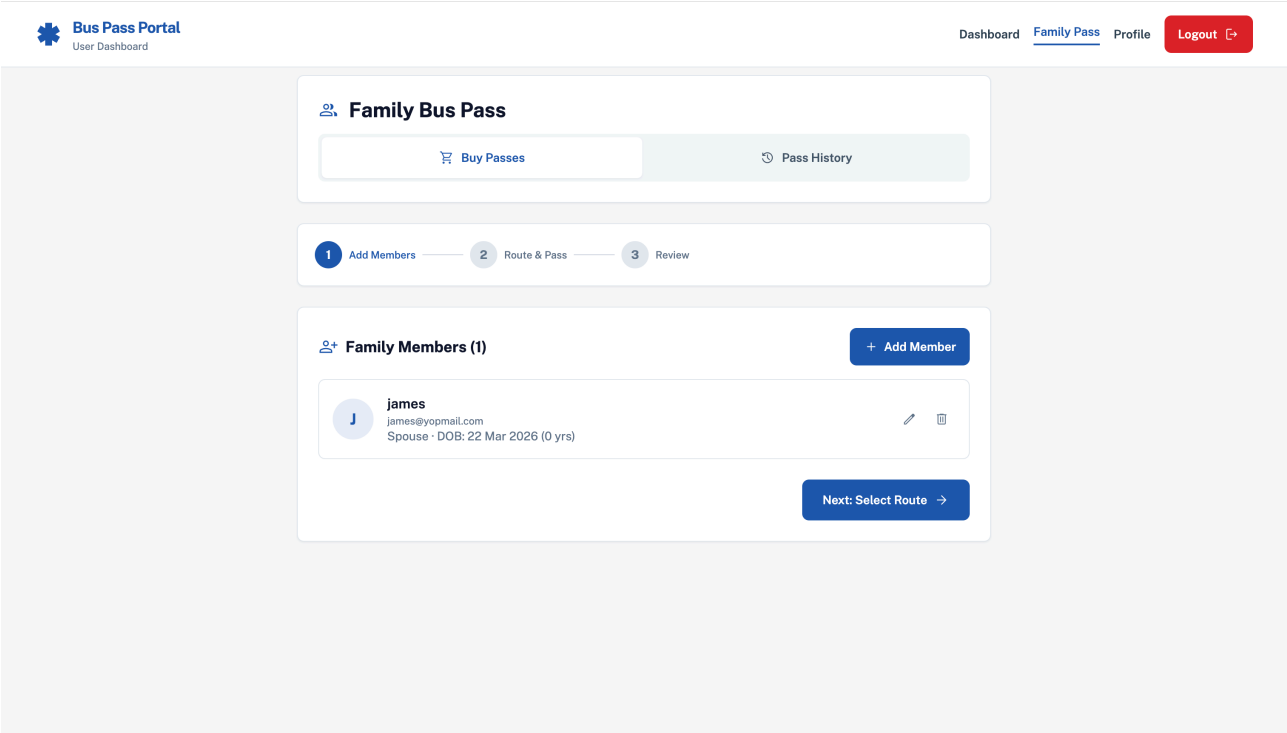

Figure: Family Pass - Member Added (Step 1) - After adding a member, the family members list shows the added member with name, email, relation, and DOB. Members can be edited or removed. URL: http://localhost:5173/family-pass

Screenshot: Family Pass - Route & Pass Selection (Step 2)

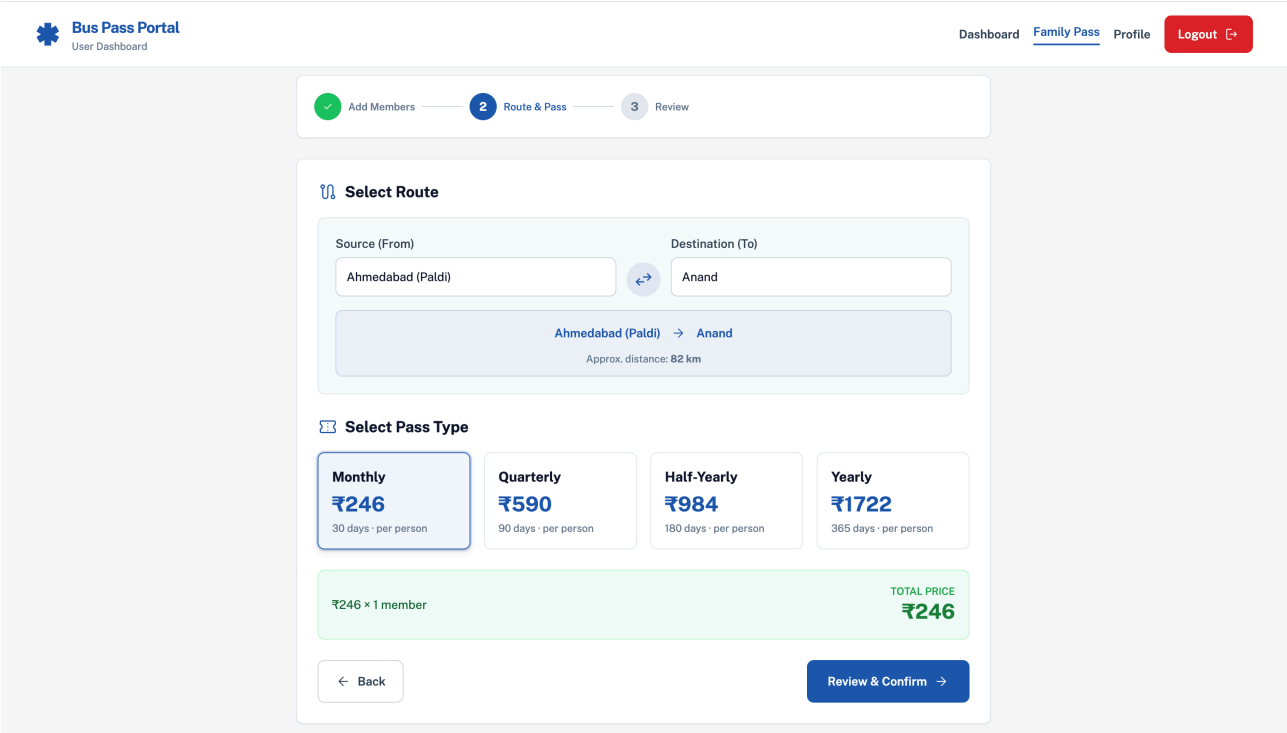

Figure: Family Pass - Route & Pass Selection (Step 2) - Step 2 allows selecting source and destination stops with fare calculation per person. Shows all four pass types with prices and total calculation. URL: http://localhost:5173/family-pass

Screenshot: Family Pass - Review & Confirm (Step 3)

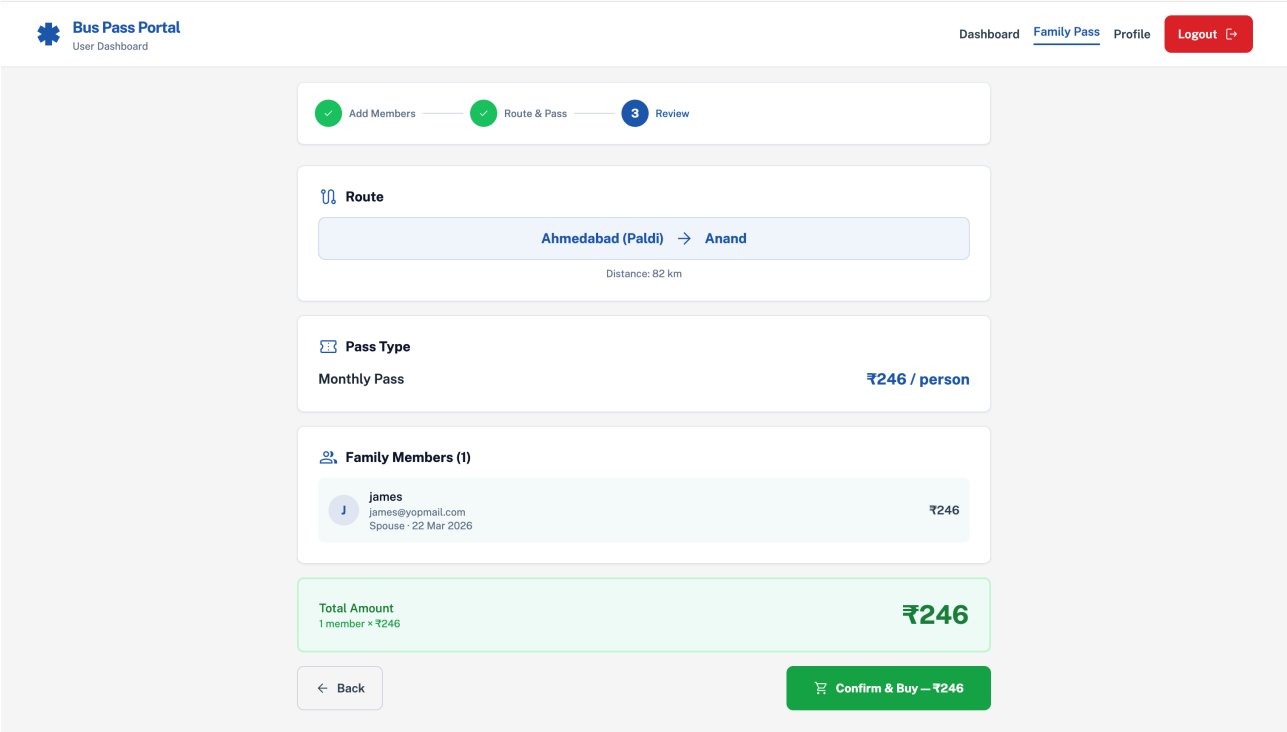

Figure: Family Pass - Review & Confirm (Step 3) - Final review step showing route details, pass type, family member list with individual pricing, and total amount with a 'Confirm & Buy' button. URL: http://localhost:5173/family-pass

Screenshot: Family Pass - Active Passes with QR Codes

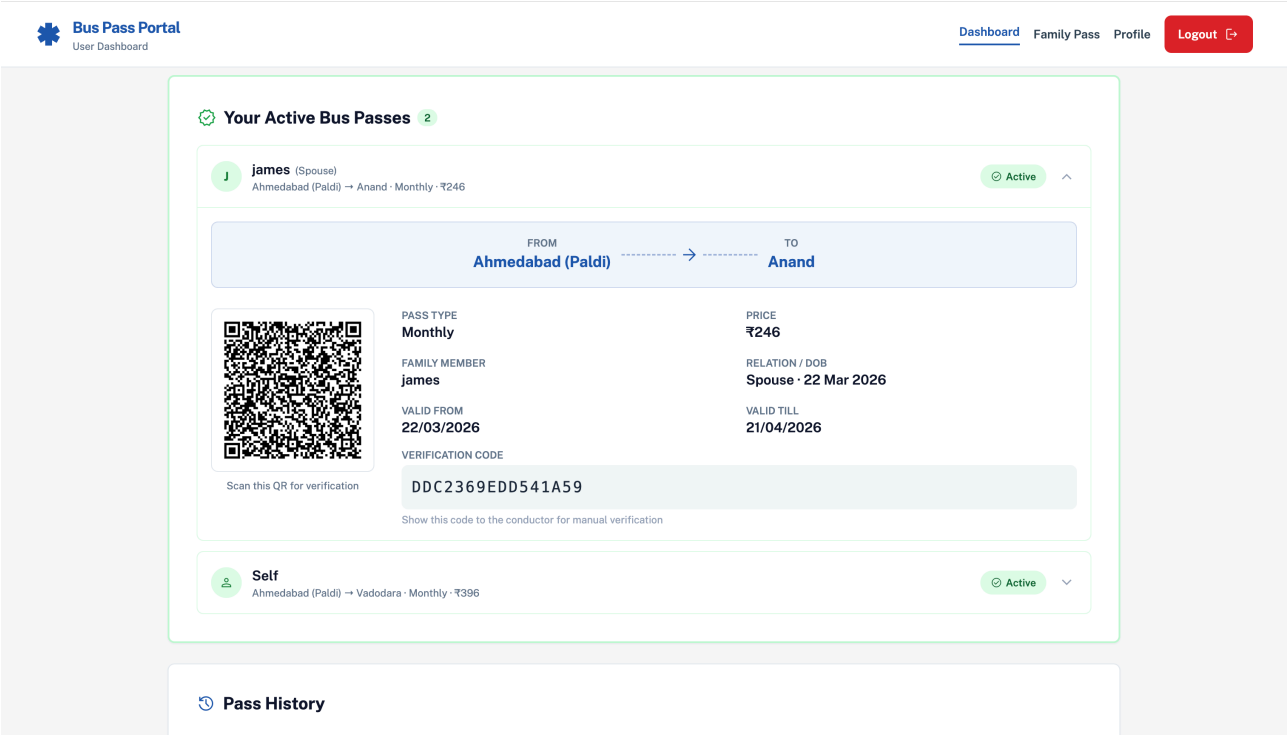

Figure: Family Pass - Active Passes with QR Codes - After approval, family passes display with QR codes, verification codes, validity dates, member details, and route information for each family member. URL: http://localhost:5173/family-pass

Screenshot: User Profile Page

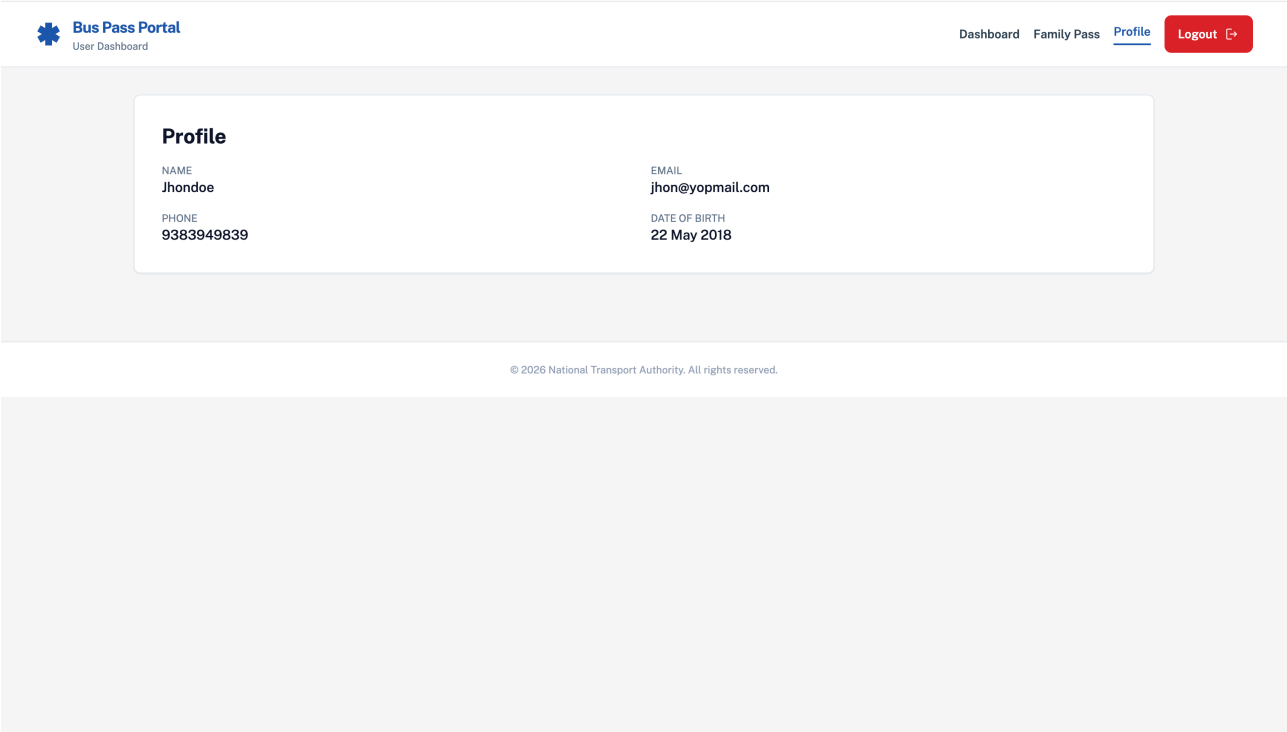

Figure: User Profile Page - Displays the user's profile information including name, email, phone number, and date of birth in a clean read-only format. URL: http://localhost:5173/profile

Screenshot: User Dashboard - Mobile View

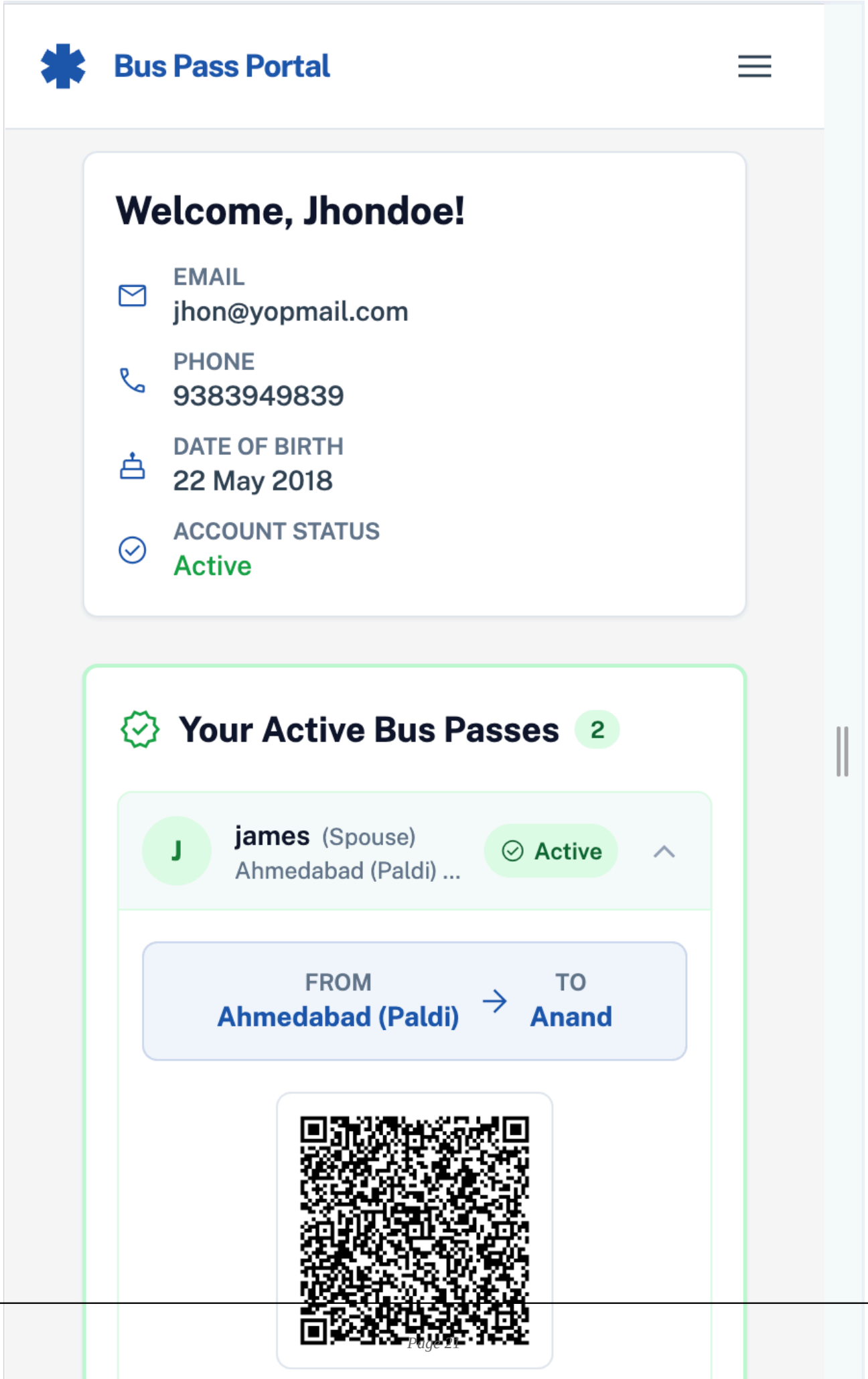
Bus Pass Portal
Welcome, Jhondoe!
EMAIL
jhon@yopmail.com
PHONE
9383949839
DATE OF BIRTH
22 May 2018
ACCOUNT STATUS
Active
Your Active Bus Passes
2
J
james (Spouse)
Ahmedabad (Paldi) ...
Active
FROM
Ahmedabad (Paldi)
TO
Anand
Scan this QR for verification

Figure: User Dashboard - Mobile View - Responsive mobile view of the user dashboard showing user details, active bus passes with QR codes, and pass information optimized for small screens. URL: http://localhost:5173/dashboard (mobile)

8. SCREENSHOTS - ADMIN SIDE

Admin Portal URL: http://localhost:5174

Production URL: https://bus-pass-management-system-1.onrender.com

Screenshot: Admin Login Page

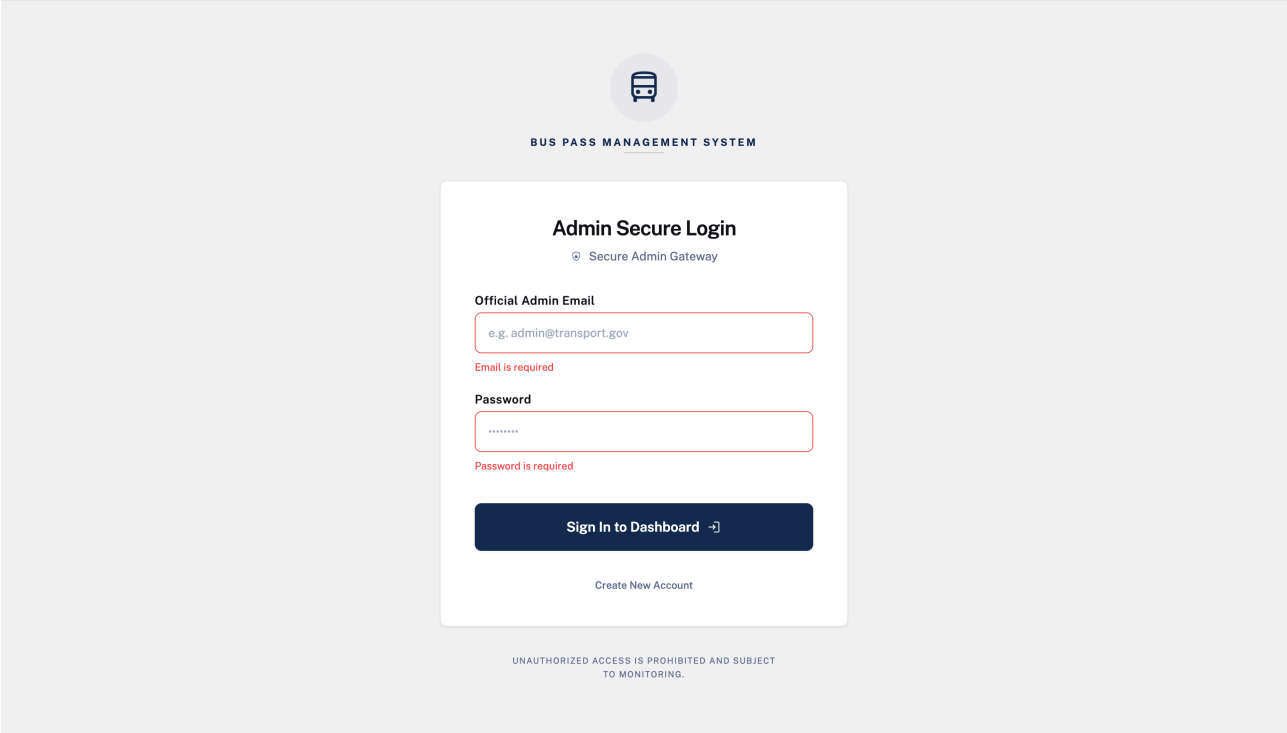

Figure: Admin Login Page - Admin secure login portal with email and password fields. Displays 'Unauthorized access is prohibited' warning. URL: http://localhost:5174/login

Screenshot: Admin Registration Page

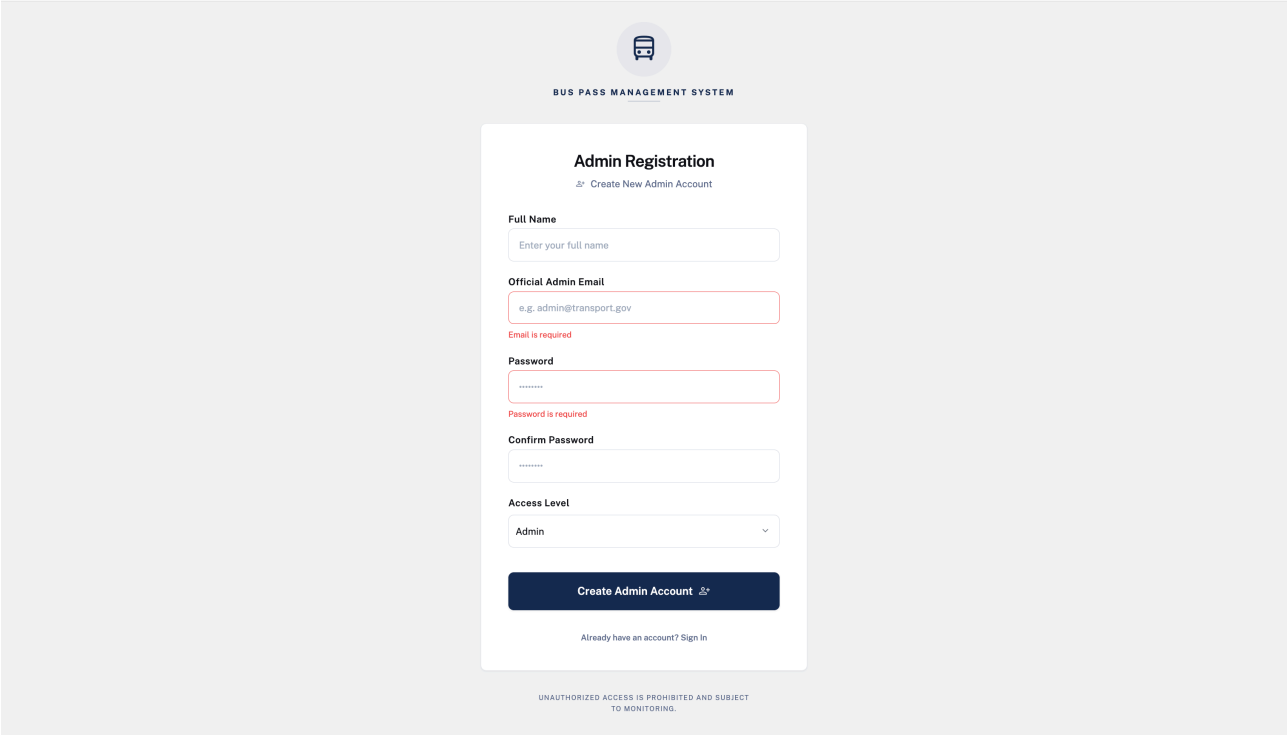

Figure: Admin Registration Page - Admin registration form with fields for Full Name, Official Admin Email, Password, Confirm Password, and Access Level dropdown (Admin, Superadmin, Conductor). URL: http://localhost:5174/register

Screenshot: Admin Dashboard - Pending Passes

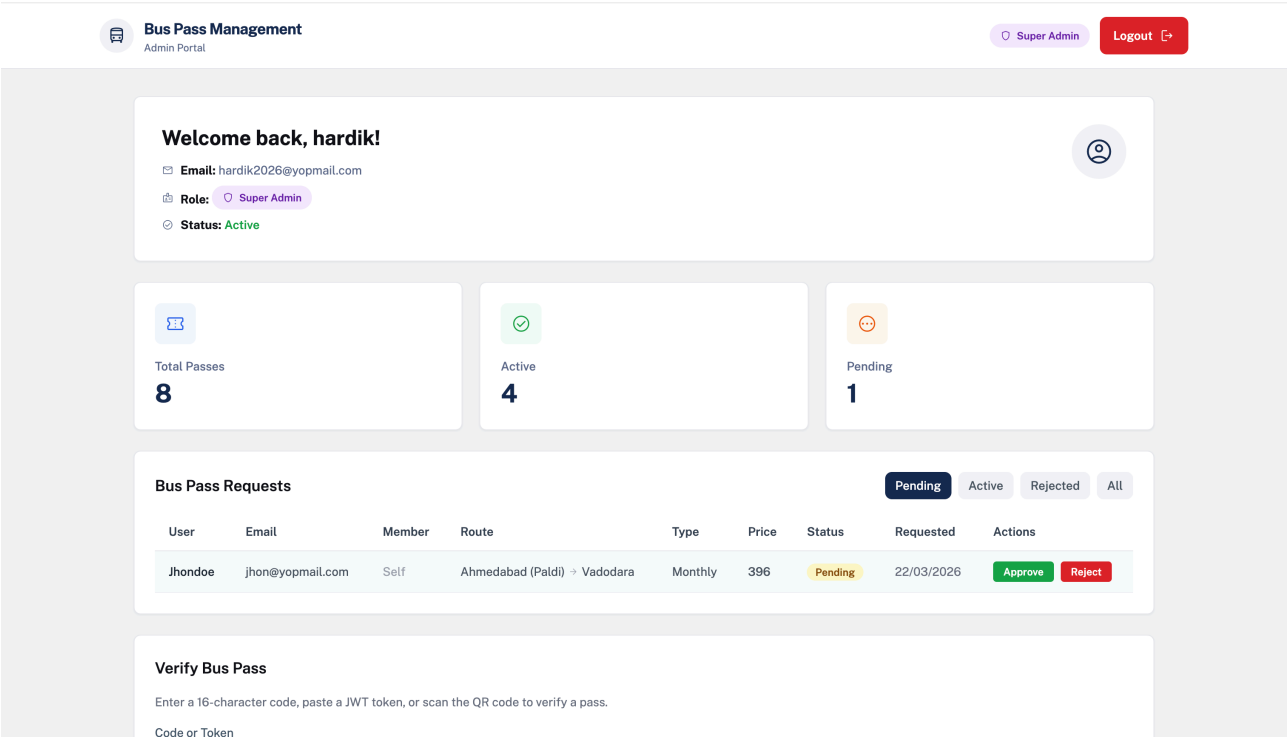

Figure: Admin Dashboard - Pending Passes - Admin dashboard showing welcome message, role badge, statistics cards (Total Passes, Active, Pending), and pending bus pass requests table with Approve/Reject action buttons. URL: http://localhost:5174/dashboard

Screenshot: Admin Dashboard - Active Passes

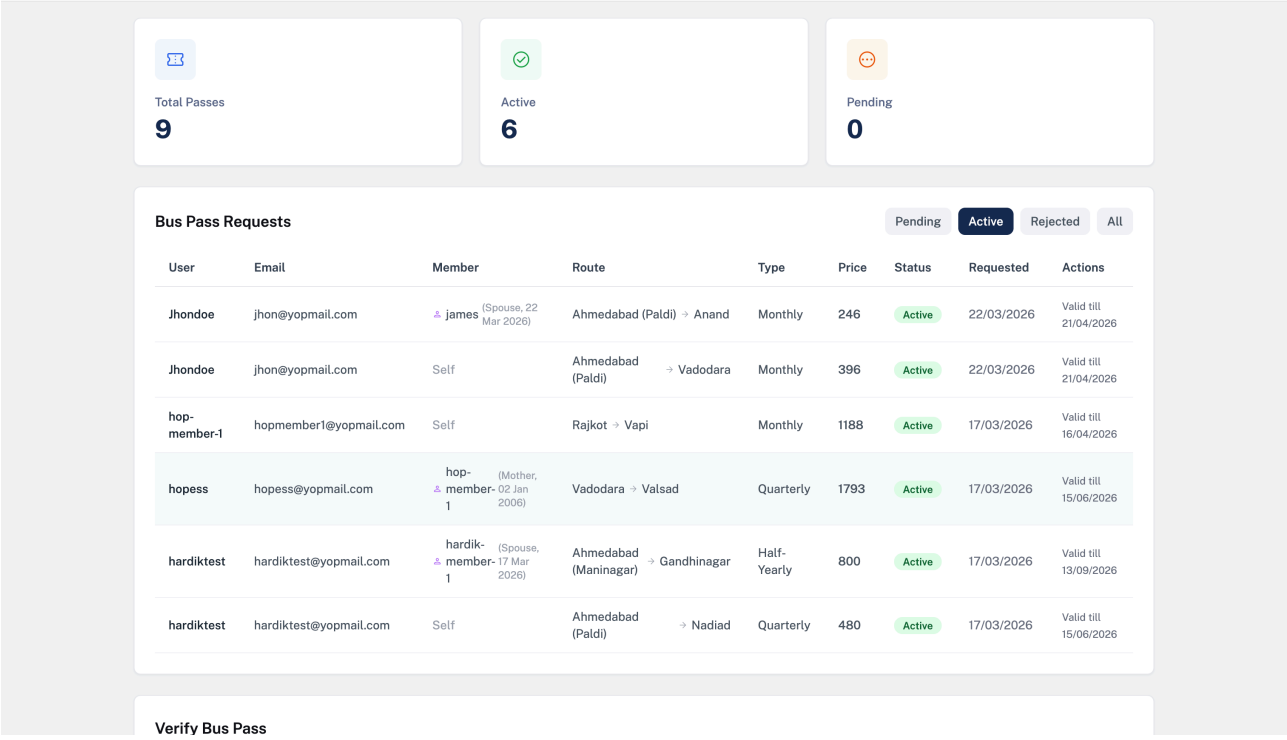

Figure: Admin Dashboard - Active Passes - Active passes tab showing all approved passes with user details, member info, route, pass type, price, status, requested date, and validity dates. URL: http://localhost:5174/dashboard

Screenshot: Admin Dashboard - Rejected Passes

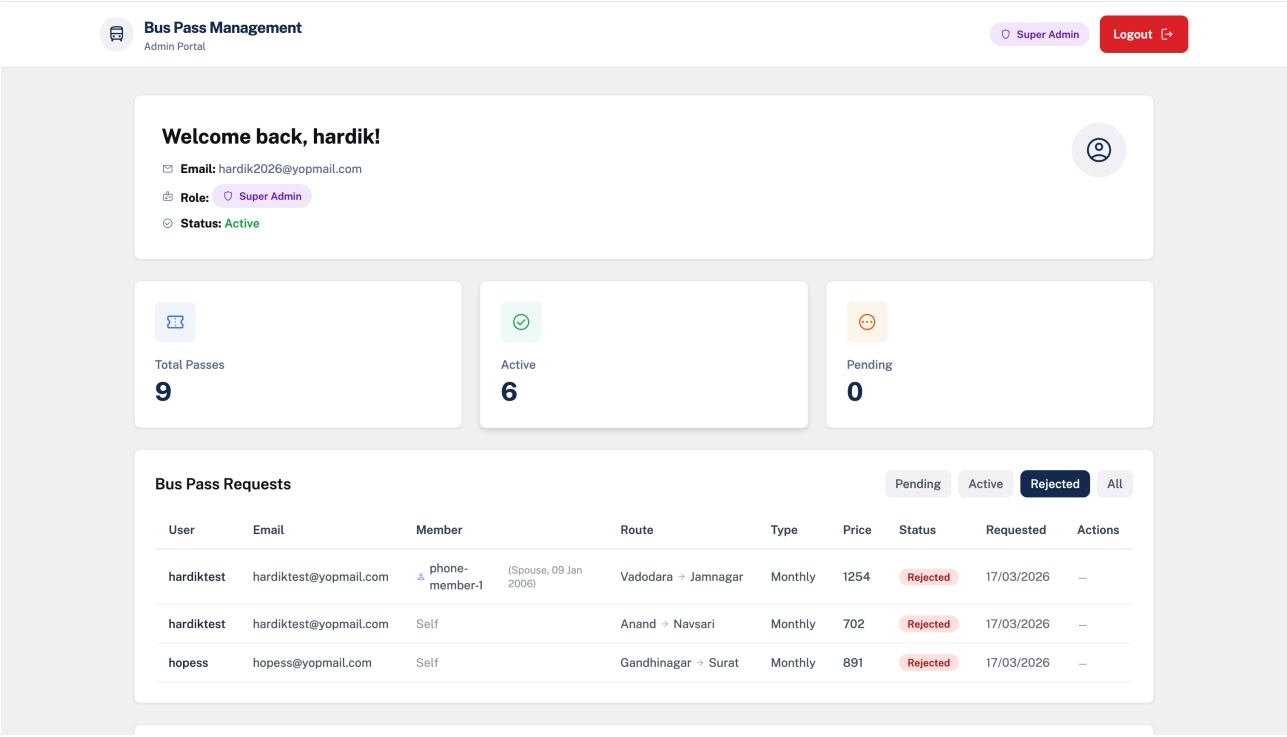

Figure: Admin Dashboard - Rejected Passes - Rejected passes tab displaying all rejected pass applications with user information, route details, and rejection status. URL: http://localhost:5174/dashboard

Screenshot: Admin Dashboard - QR Code Scanner & Manual Verification

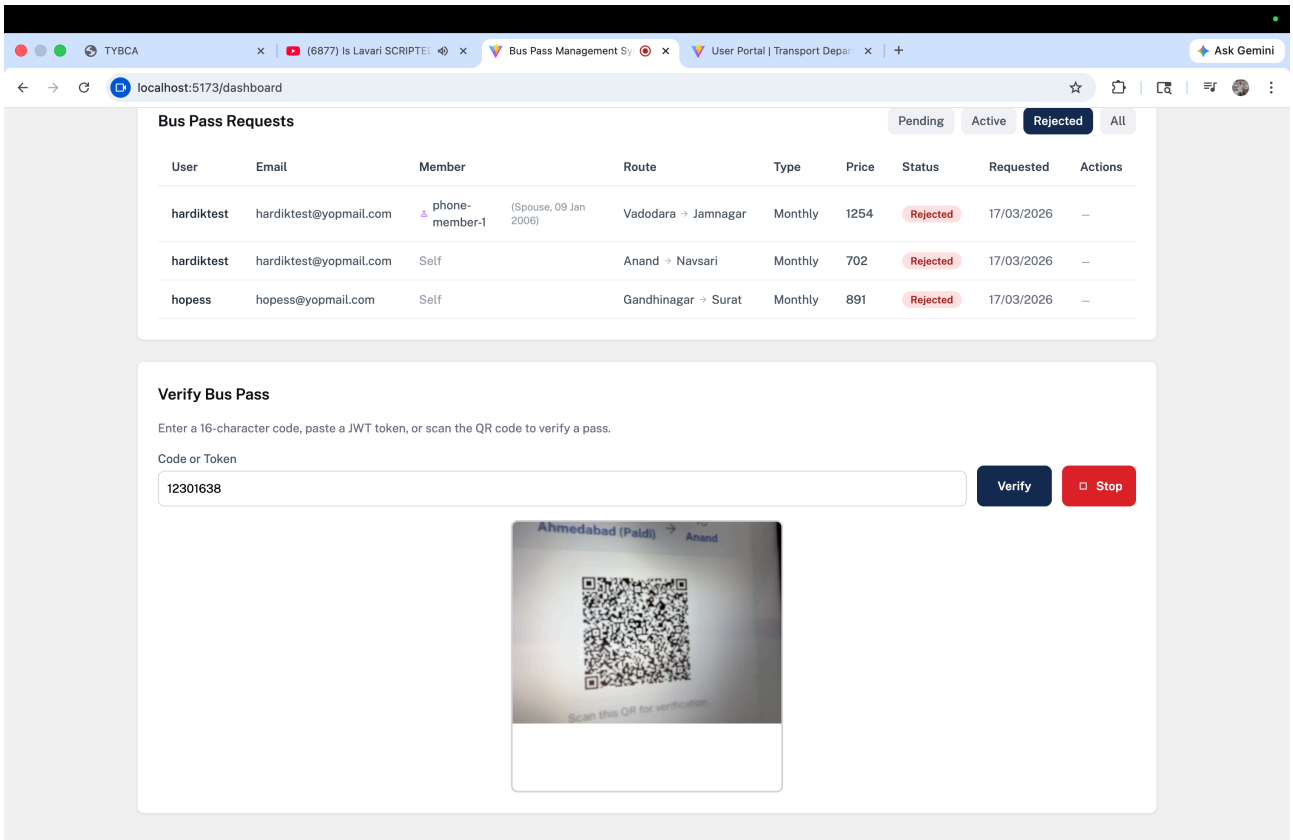

Figure: Admin Dashboard - QR Code Scanner & Manual Verification - The verify bus pass section with a text input for 16-character code or JWT token, Verify button, and integrated camera-based QR code scanner for real-time pass scanning. URL: http://localhost:5174/dashboard

Screenshot: Conductor Mode - Pass Verification Result

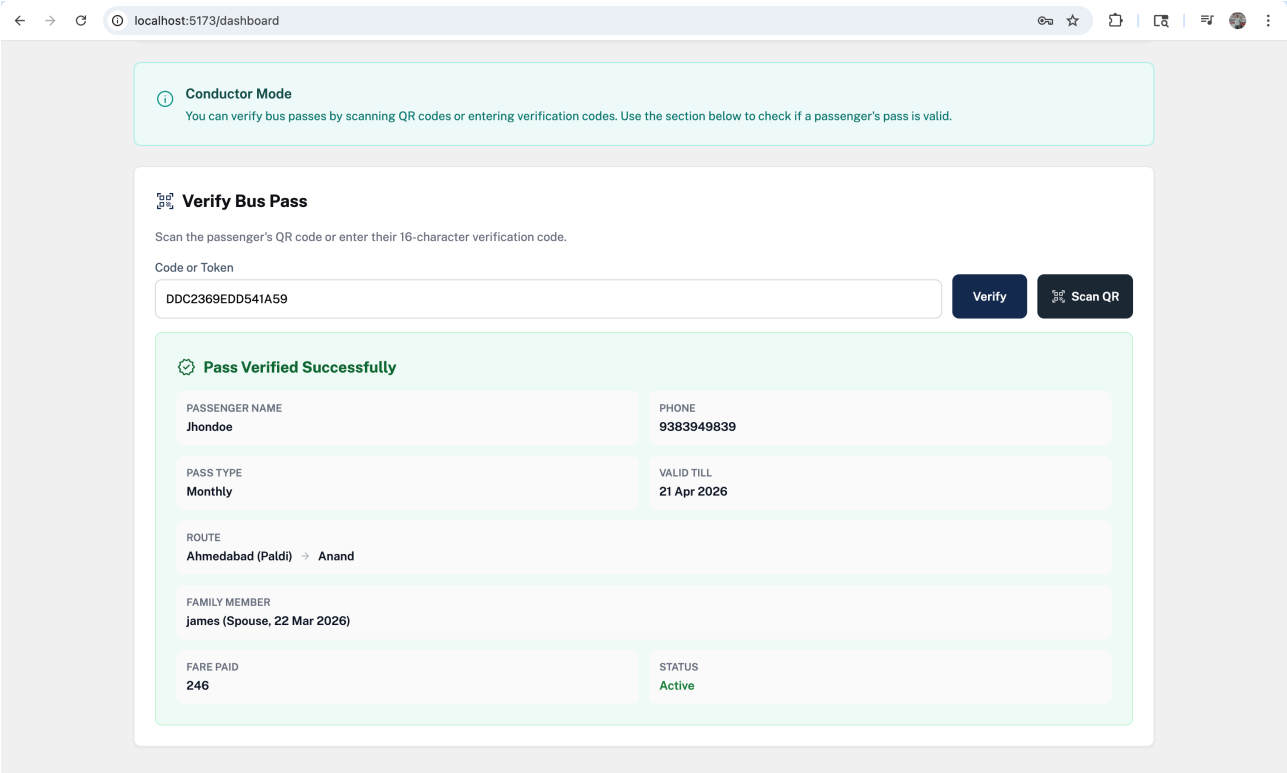

Figure: Conductor Mode - Pass Verification Result - Conductor verification view showing 'Pass Verified Successfully' with passenger name, phone, pass type, validity, route, family member details, fare paid, and active status. URL: http://localhost:5173/dashboard (conductor)

9. CONCLUSION

The Bus Pass Management System successfully demonstrates a full-stack web application built using the MERN stack (MongoDB, Express.js, React.js, Node.js). The system provides a complete digital solution for bus pass management with the following key achievements:

- Digitized the entire bus pass lifecycle from purchase to verification.
- Implemented secure JWT authentication with bcrypt password hashing and role-based access control.
- Built intelligent fare calculation using the Haversine formula across 20 Gujarat bus stops.
- Developed QR code-based verification for real-time pass validation by conductors.
- Created family pass feature with bulk purchasing and auto account creation.
- Designed responsive UI with Tailwind CSS for desktop and mobile devices.
- Deployed on Render.com with separate user portal, admin portal, and backend API.

The project demonstrates proficiency in modern web development including RESTful API design, NoSQL database modeling, token-based authentication, QR code generation/scanning, and responsive UI design.